

Intro to Reinforcement Learning

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What is Machine Learning?

Hot Take: All of AI is About Utility Maximization

1. **Solve A Math Problem**
2. **Predict A Binding Affinity that is Close in a Given Metric**
3. **Have a Robot Execute a Task to high tolernace**
4. **Generate one or many appealing images**
5. **Produce a text transcript of an audio recording**

What is Machine Learning?

Hot Take: All of AI is About Utility Maximization

1. We have an AI model p_θ

2. We are given **inputs** x

3. We provide **outputs** y

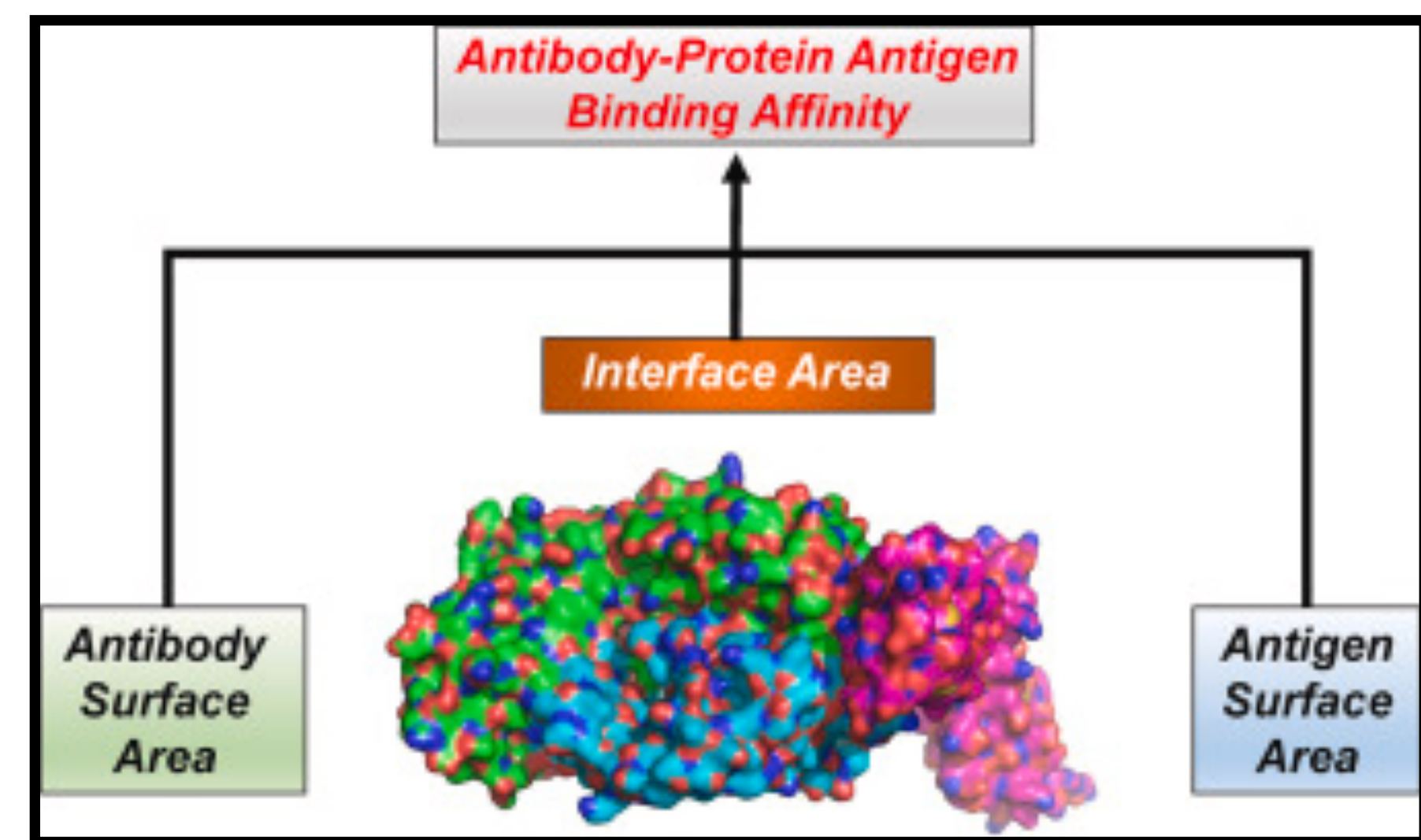
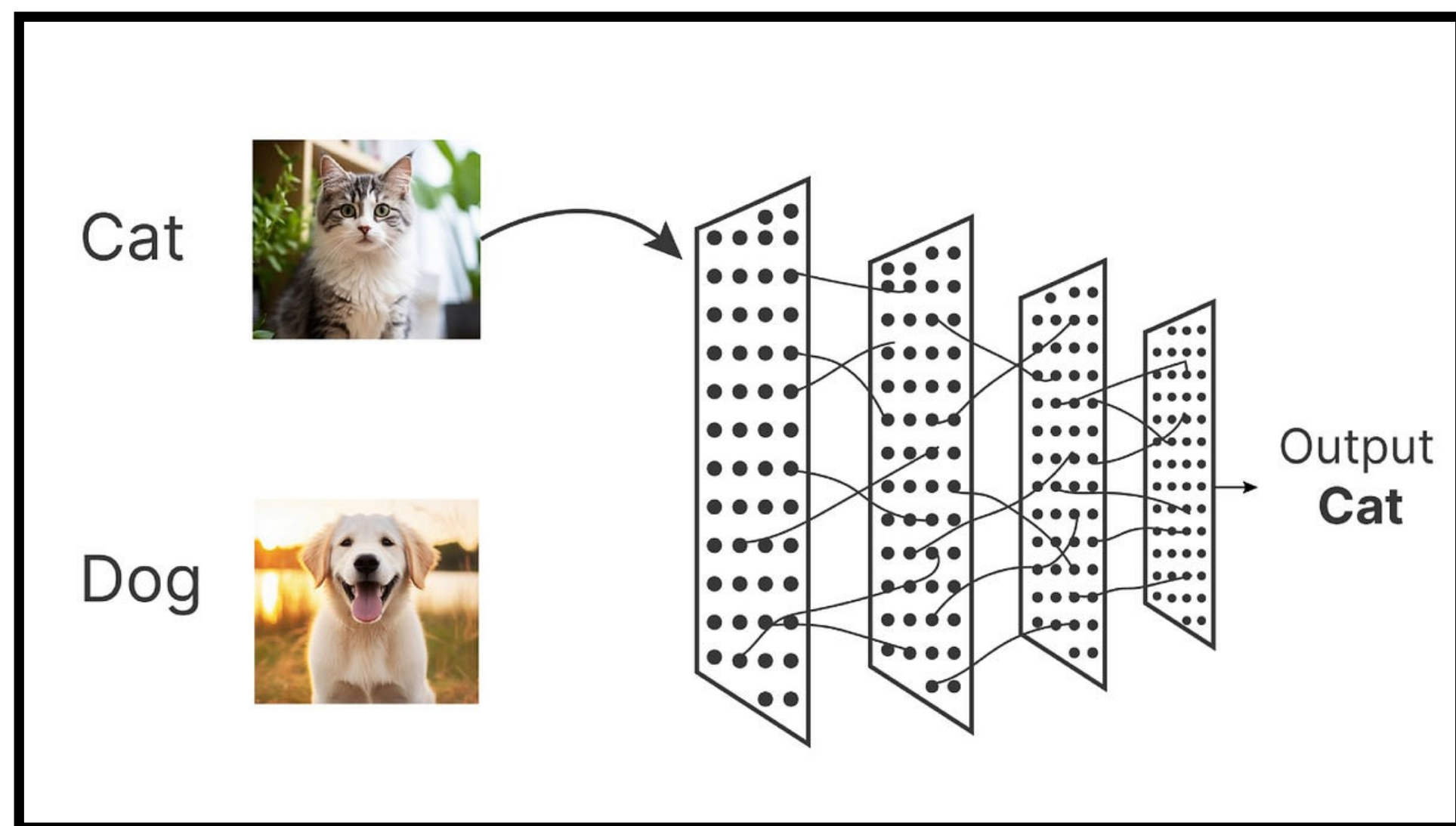
$$\max_{\theta} \mathbb{E}_x \text{Utility}(y \sim p_\theta(\cdot | x), x)$$

Supervised Learning

$$\max_{\theta} \mathbb{E}_x \text{Utility}(\mathbf{y} \sim p_{\theta}(\cdot | \mathbf{x}), \mathbf{x})$$

Supervised Learning:

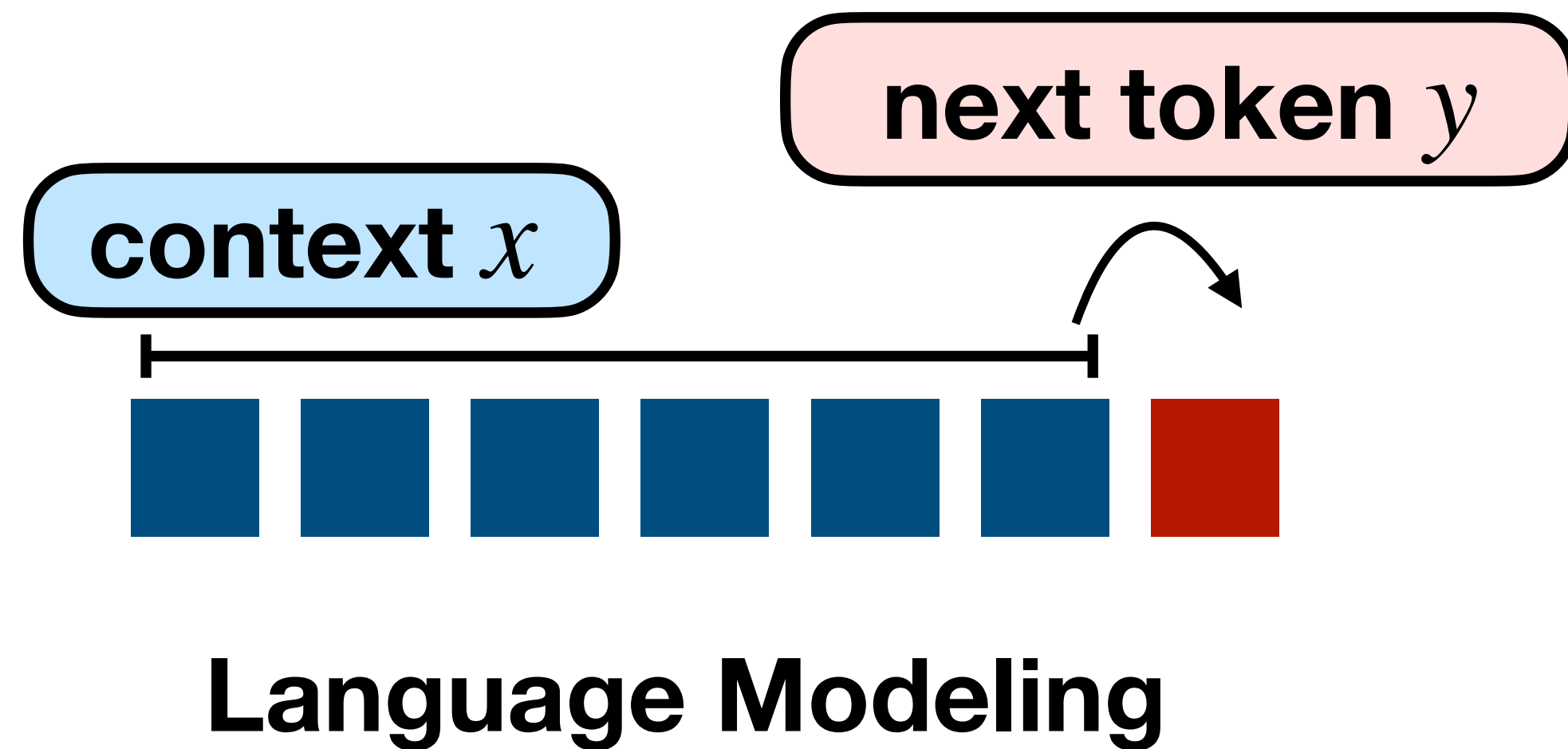
$$\text{Utility} = - \mathbb{E}_{\mathbf{y} \sim p_{\theta}(\cdot | \mathbf{x})} \text{loss}(\mathbf{y}; \mathbf{y}^{\star}(\mathbf{x}))$$



Generative Modeling

$$\text{Utility} = -1 \cdot \text{dist}(p_{\text{data}}(\cdot | x), p_{\theta}(\cdot | x))$$

We want to fit a **distribution**



We maximize log-probability

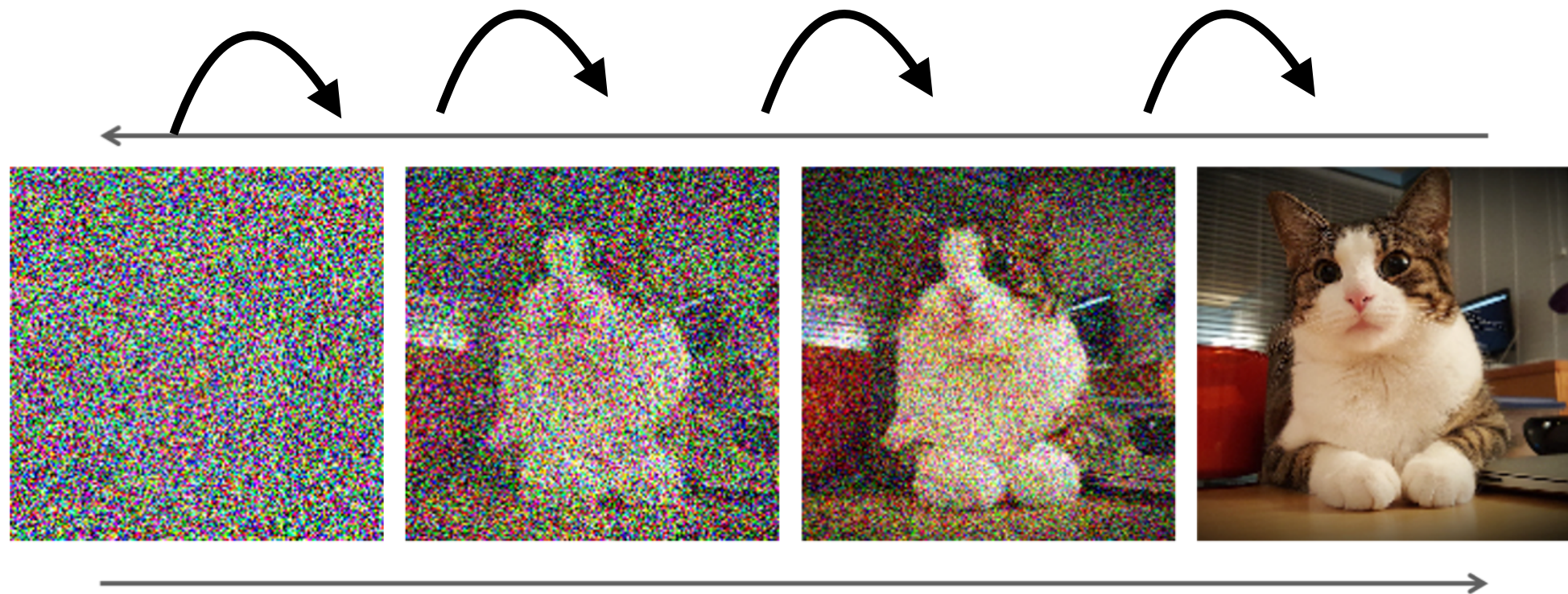
$$\max_{\theta} \mathbb{E}_x \mathbb{E}_{y \sim \text{data}} [\log p_{\theta}(y | x)]$$

$$\min_{\theta} \mathbb{E}_x \mathbb{E}_{y \sim \text{data}} \text{KL}(p_{\text{data}}(\cdot | x) \parallel p_{\theta}(\cdot | x))$$

Generative Modeling

$$\text{Utility} = -1 \cdot \text{dist}(p_{\text{data}}(\cdot | x), p_{\theta}(\cdot | x))$$

We want to fit a **distribution**



Learn to sample a “**good image**”

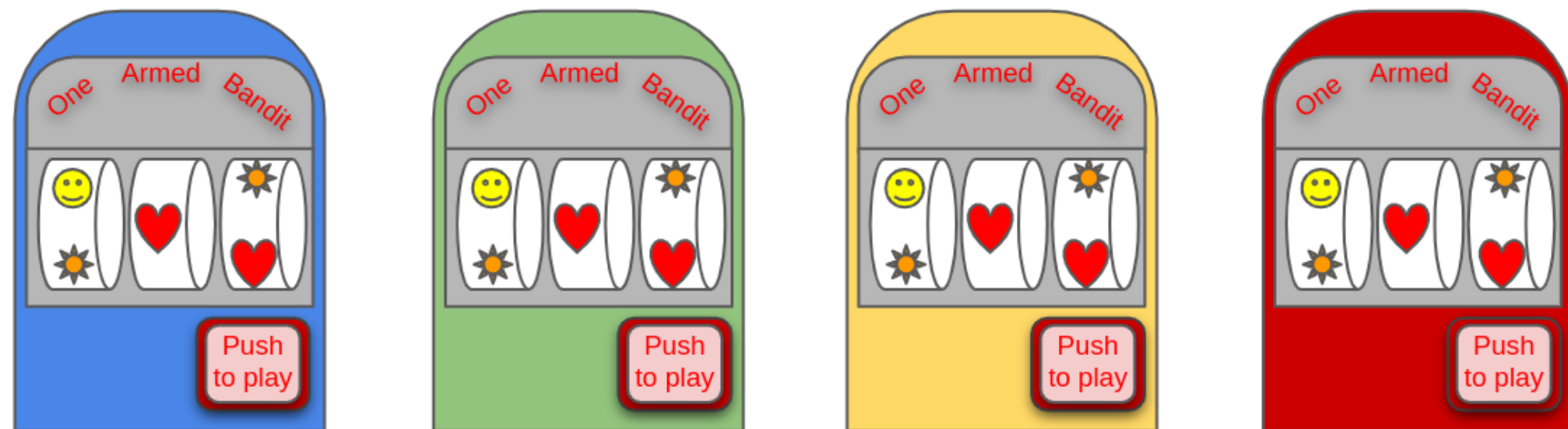
We maximize “some other matching”

$$\max_{\theta} \mathbb{E}_x \mathbb{E}_{y \sim \text{data}} [\text{"flow matching loss"}]$$

Decision Making

$$\text{Utility} = \mathbb{E}_{a \sim p_{\theta}}[r(a, x)]$$

We want optimize over **our distribution**



x = **name of casino** (context)

a = **which slot machine** (action)

Contextual Bandits

‘classification with **reward** semantics’

you don’t care about **directly predicting** the reward, only taking the **right action**

Multi-Step Decision Making (RL)

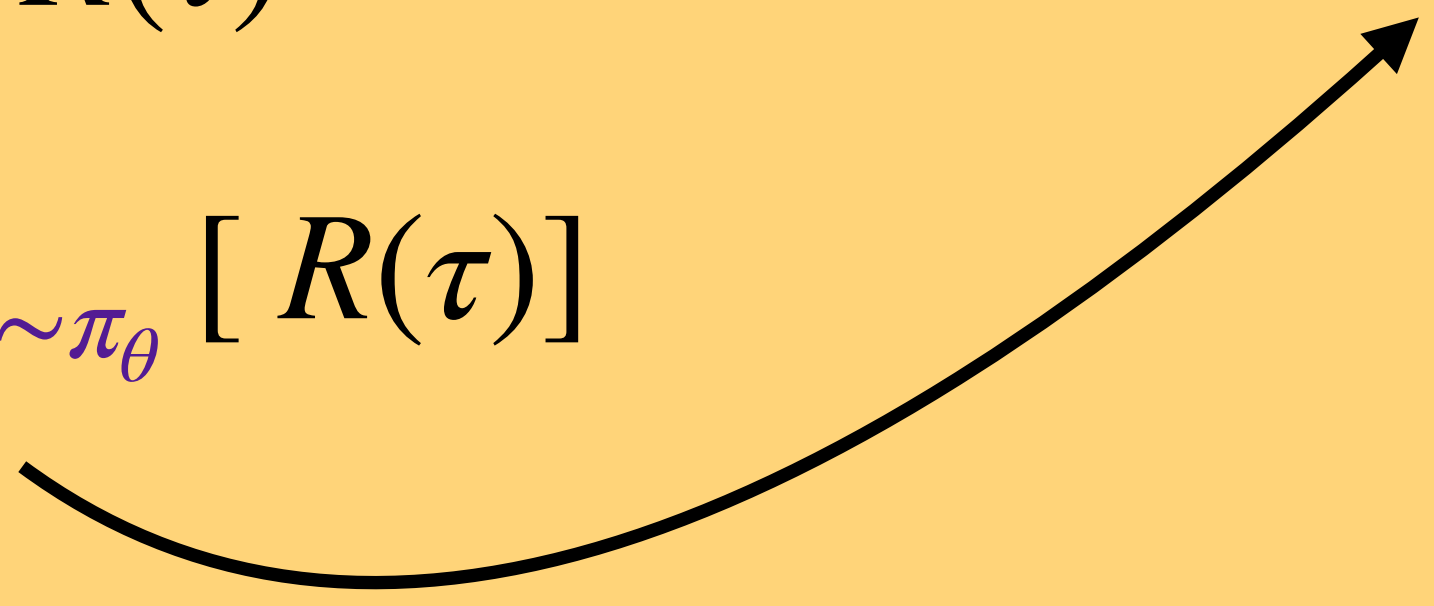
1. Policy π_θ and dynamics p induce trajectories $\tau = (s_1, a_1, s_2, a_2, \dots)$

Note: Trajectories Depend on Both Learner Actions and Dynamics Randomness

2. We receive reward $R(\tau)$

3. We aim for $\max_{\theta} \mathbb{E}_{\tau \sim \pi_\theta} [R(\tau)]$

**Learn over unknown
rewards or dynamics**



Multi-Step Decision Making (RL)

1. Policy π_θ and dynamics p induce trajectories $\tau = (s_1, a_1, s_2, a_2, \dots)$

**Remark: Bandits + Classification
are special cases.**

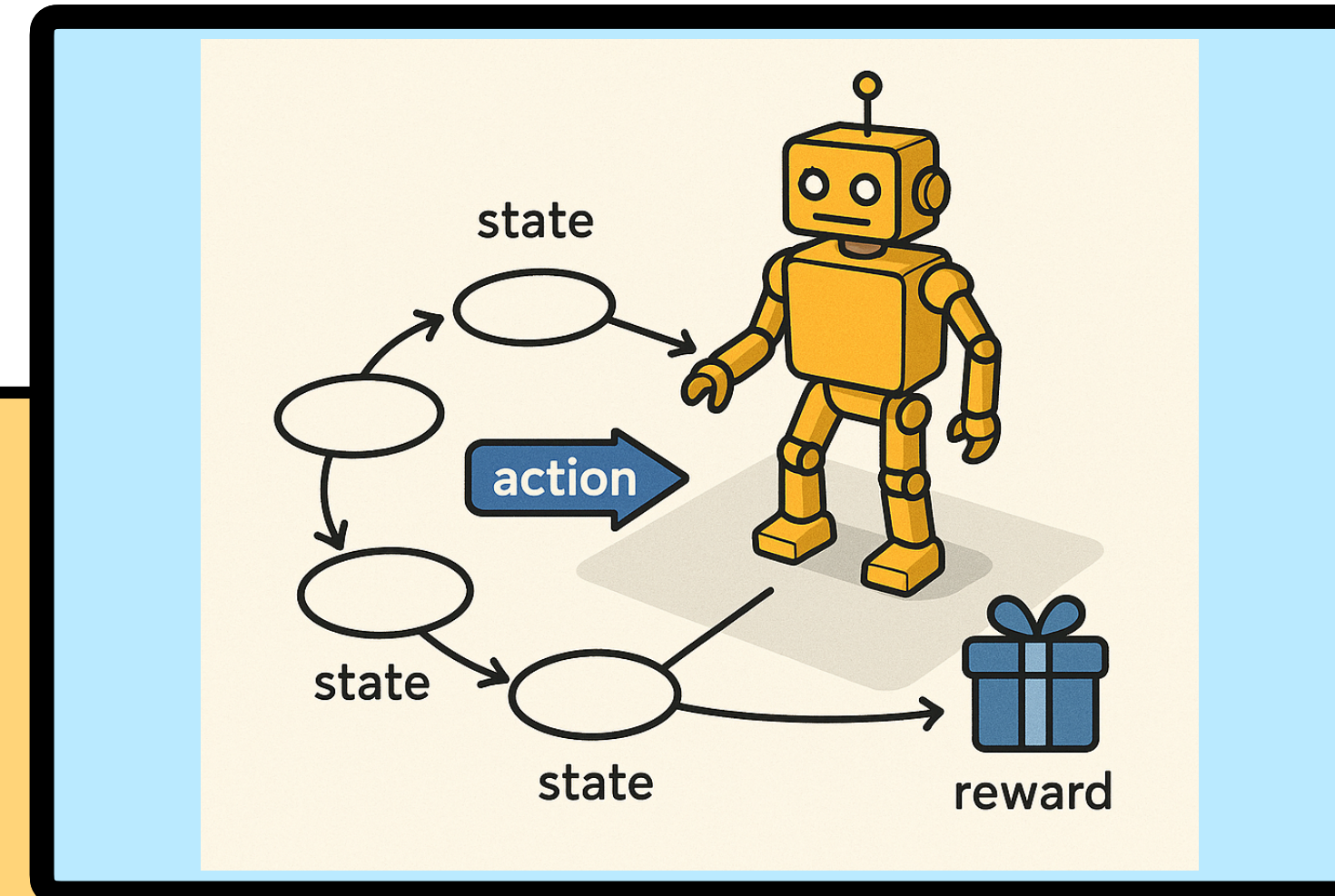
Multi-Step Decision Making (RL)

1. Policy π_θ and dynamics p induce trajectories $\tau = (s_1, a_1, s_2, a_2, \dots)$

Special Case: We say the MDP is Markov if $p(s_{t+1} \mid s_t, a_t)$

$$R(\tau) = \sum_t R_t(s_t, a_t)$$

Will show: optimal policy
Markov $\pi(\cdot \mid s_t)$

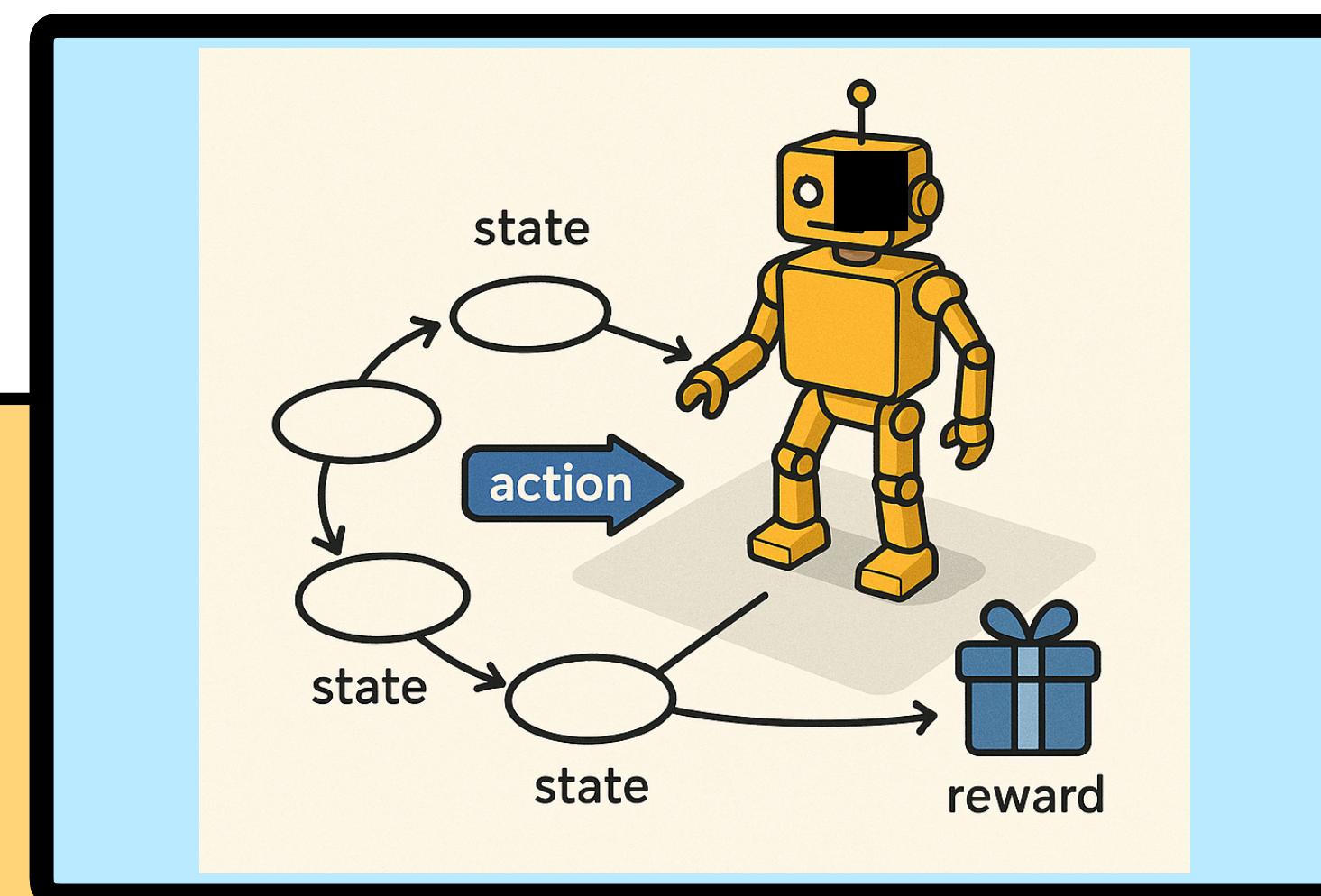


Multi-Step Decision Making (RL)

1. Policy π_θ and dynamics p induce trajectories $\tau = (s_1, a_1, s_2, a_2, \dots)$

Special Case: We say the Problem is PODMP if the policy can only see $o_t \sim p_{\text{obs}}(s_t)$

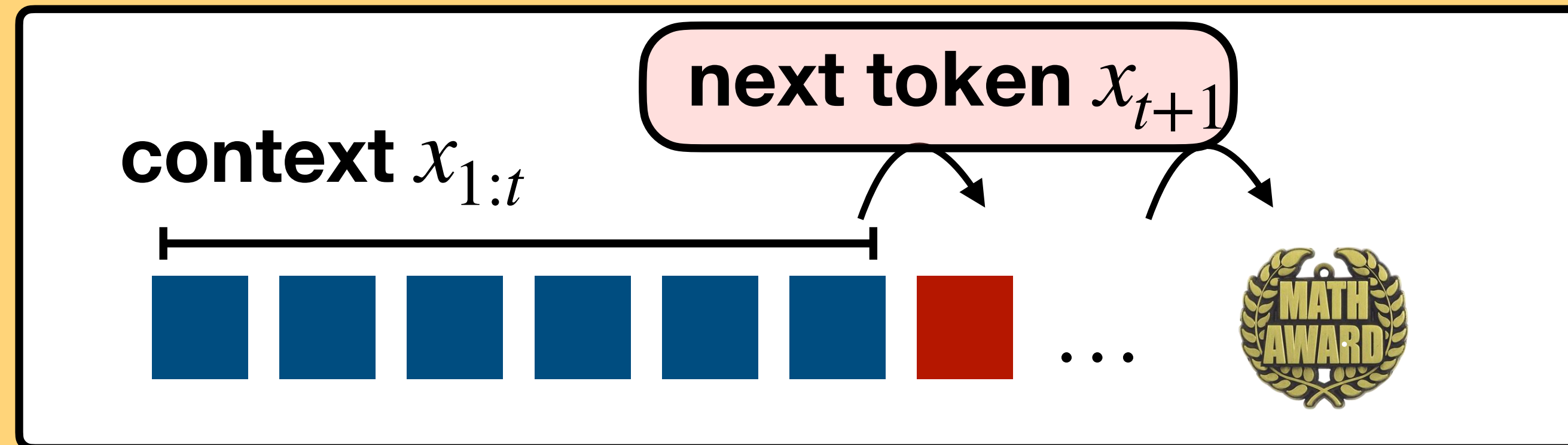
Will show: optimal policy needs history $\pi(\cdot \mid o_{1:t}, a_{1:t})$



Multi-Step Decision Making (RL)

1. Policy π_θ and dynamics p induce trajectories $\tau = (s_1, a_1, s_2, a_2, \dots)$

Special Case: We say the problem is an auto-regressive MDP if a_t are tokens $x \in \mathbb{X}$ and $s_{t+1} = \text{concat}(s_t, a_t)$



Multi-Step Decision Making v.s Generative Modeling

$$\max_{\theta} \mathbb{E}_{\tau \sim \pi_{\theta}} [R(\tau)]$$

Maximize Reward Under **Own Distribution**

$$\max_{\theta} \mathbb{E}_{\tau \sim \text{data}} [\log p_{\theta}(\tau)]$$

Maximize Likelihood Under **Data Distribution**

**Data Distribution v.s.
Own Distribution**

**Rewards v.
Likelihoods**

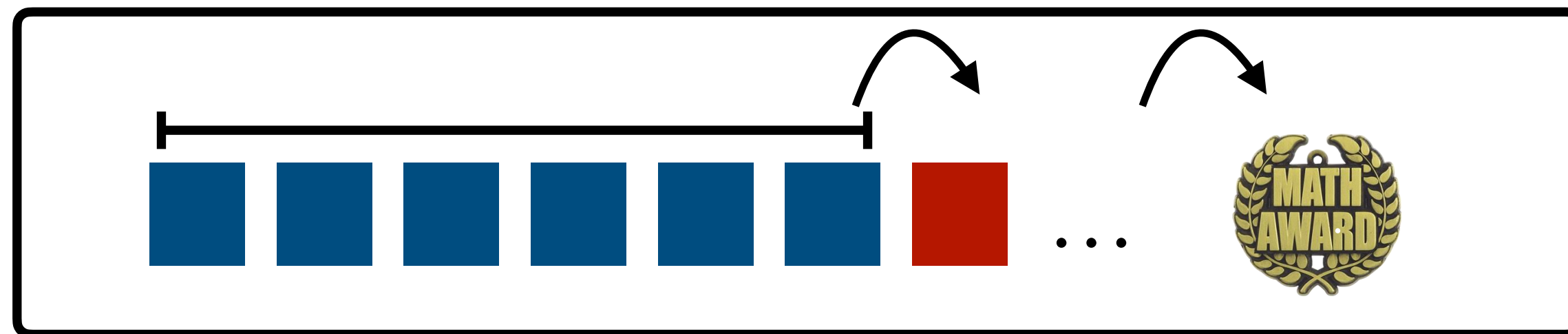
Multi-Step Decision Making v.s Generative Modeling

$$\max_{\theta} \mathbb{E}_{\tau \sim \pi_{\theta}} [R(\tau)]$$

Reward of Sequence

$$\max_{\theta} \mathbb{E}_{\tau \sim \text{data}} [\log p_{\theta}(\tau)]$$

Likelihood of Sequence



Example: Auto-regressive MDPs

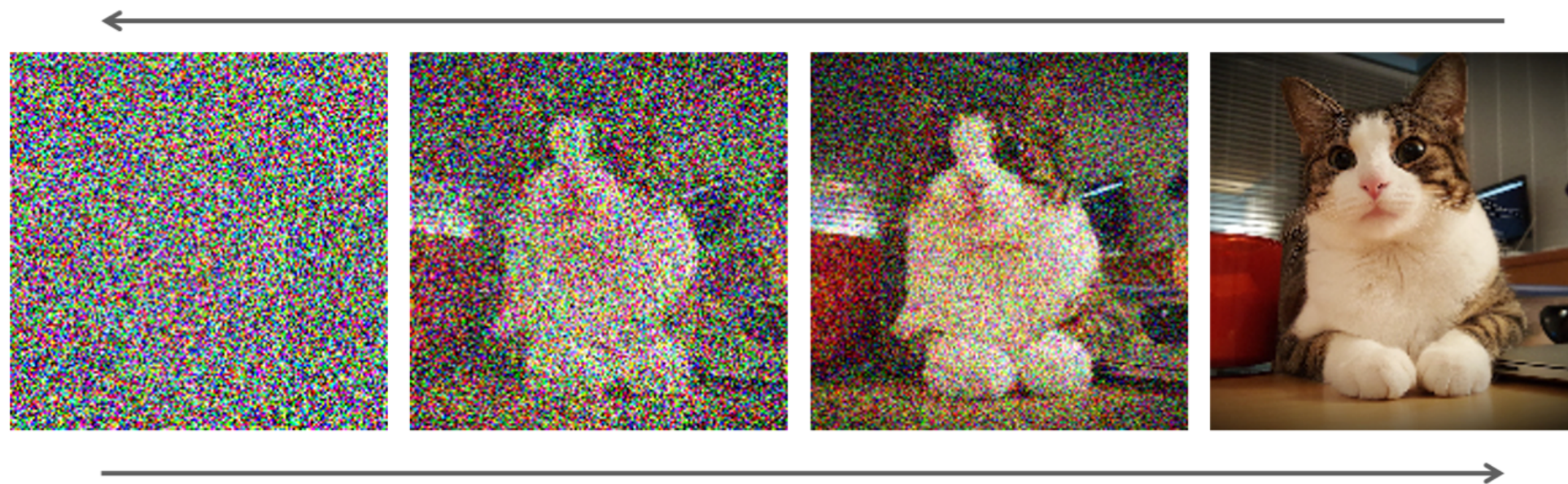
Multi-Step Decision Making v.s Generation Modeling

$$\max_{\theta} \mathbb{E}_{\tau \sim \pi_{\theta}} [R(\tau)]$$

Reward of Sequence

$$\max_{\theta} \mathbb{E}_{\tau \sim \text{data}} [\text{flow loss} (p_{\theta}, \tau)]$$

Likelihood of Sequence

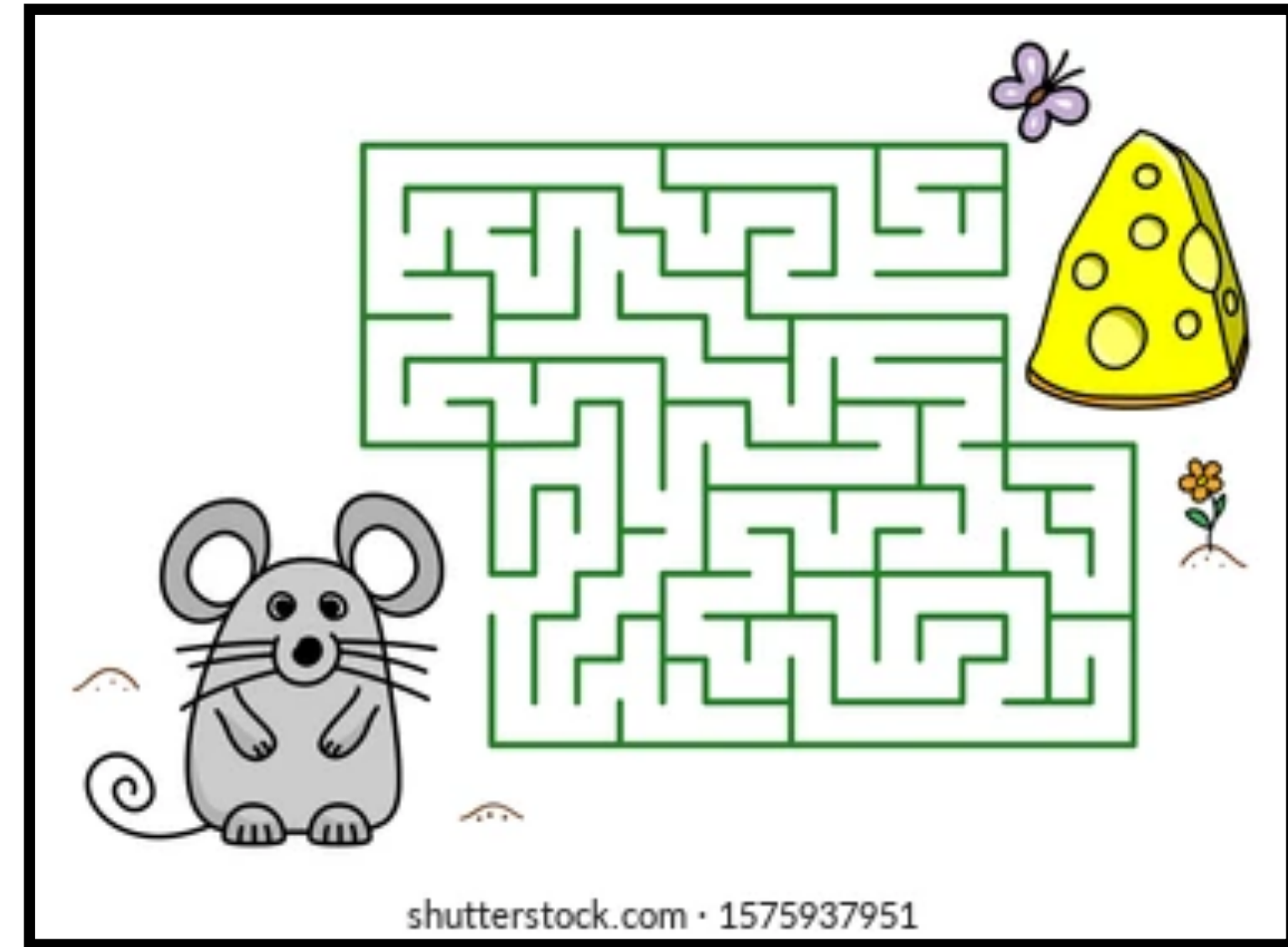


Controversial Example: Diffusion Models

Why is RL challenging?

$$\max_{\theta} \mathbb{E}_{\tau \sim \pi_{\theta}} [R(\tau)]$$

Challenge 1: We need to search
(**exploration**)



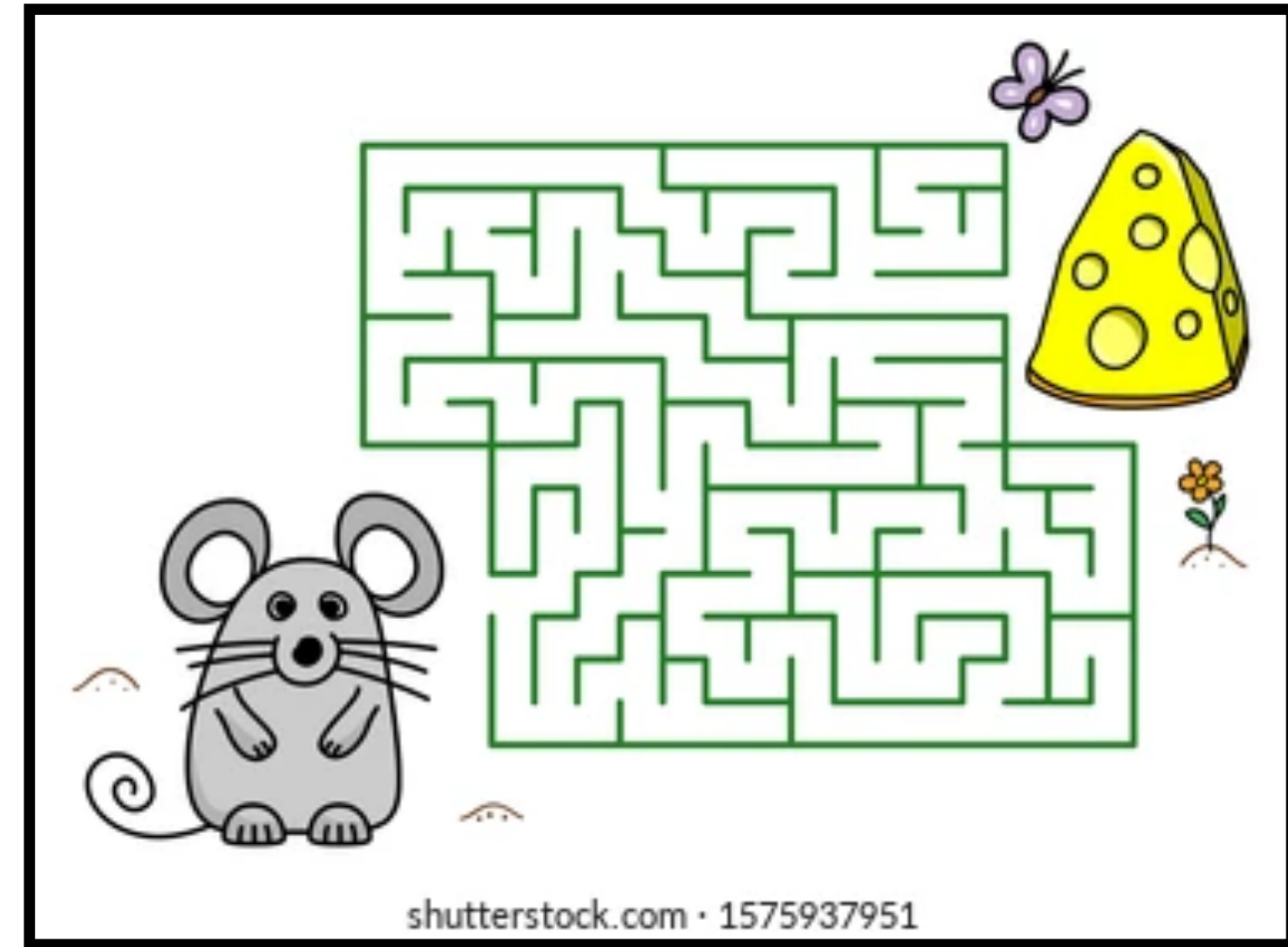
If $R(\tau) = \mathbb{I}(\text{found cheese})$, then I need to be able to have some success before I can hope to learn anything

*Must also avoid **exploration collapse***

Why is RL challenging?

$$\max_{\theta} \mathbb{E}_{\tau \sim \pi_{\theta}} [R(\tau)]$$

Challenge 2: Learning what caused success (**credit assignment**)



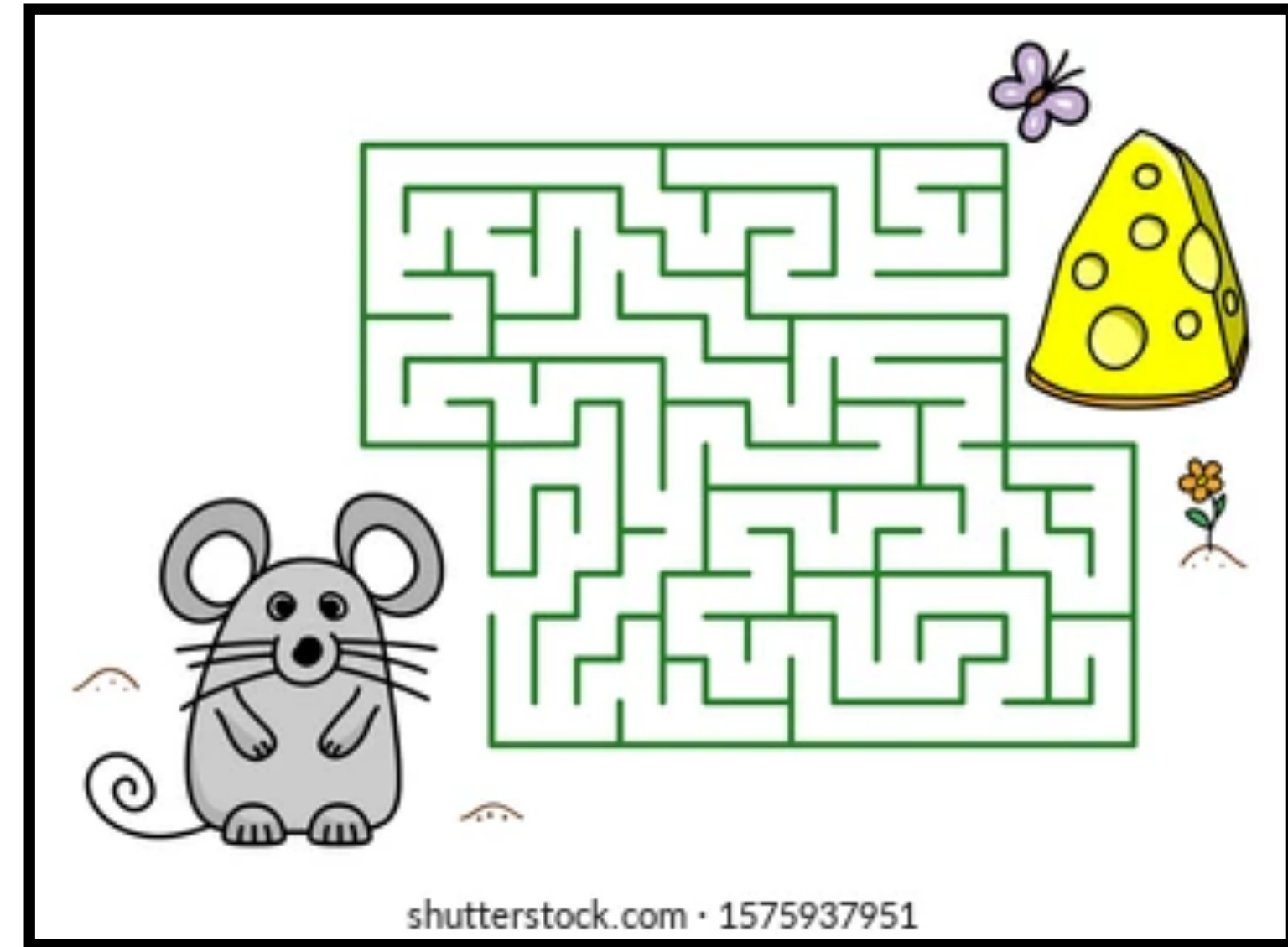
If the mouse fails, how does it identify which was its **wrong turn**?

Manifests as *gradient variance*

Why is RL challenging?

$$\max_{\theta} \mathbb{E}_{\tau \sim \pi_{\theta}} [R(\tau)]$$

Challenge 3: Learning distribution is changing (**non-stationarity**)



As the mouse learns, its policy distribution changes, making it harder to naively learn from **past behavior**.

Why is RL challenging?

$$\max_{\theta} \mathbb{E}_{\tau \sim \pi_{\theta}} [R(\tau)]$$

Challenge 1: We need to search (exploration**)**

(Warm-up from supervised learning, curriculum, still hard)

Challenge 2: Learning what caused success (credit assignment**)**

(Critic learning, rubrics, and other things)

Challenge 3: Learning distribution is changing (non-stationarity**)**

*(Generally challenging, but uses **off-policy learning**)*

Organization of the Course

Max is lazy so we ~~will go~~ went to the website.